

Part 1:

Q1: A student's eligibility to sit for an exam depends on their attendance, which must be at least 90%. Take the following information from the user:

- Number of classes held.
- The number of classes attended.

Then calculate the percentage of class attended and print:

- Percentage of classes attended.

Whether the student is permitted to take the exam or not. (5 points)

Q2: Write a program to verify if the number entered by the user is a three-digit number and print a corresponding message. (5 points)

Part 2: Python Quiz Application (20 points)

Quiz Details:

Now let's use the programming concepts that you have learned so far to create a basic Python Quiz (Multiple choice questions) application. The main functionality of the application is to ask the user various questions, collect the answers given by the user and check if the entered answers are correct or not. The application will end up displaying the final results.

Concepts that you will use here:

- Basics of python
- Control structure

Instructions for creating a quiz:

1. Give a meaningful name to your quiz application. Example: Python_Quiz_Application.
2. Enter some description for the application at the beginning of the program as comments so that users can easily understand your application
3. Ask for the name of the user that would be displayed as a player name
4. Your quiz should have three different levels (Easy, Average, and Hard). Ask the user to enter the choice and store this into a variable. Here you will use decision structures to switch between different levels.
5. Each level should have at least 5 questions with 4 options. Here you will use input(), and print() functions to display the questions. Further, you will use decision structures to check which option is correct. You can take a variable option to store the option entered by the user that can be checked using decision structures.

Note: If the user enters a wrong answer, display an “invalid choice” message to the user.

6. The application should have a calculateResult variable that will be incremented by 1 every time the user enters the correct answer for the question. However, there is no negative marking for any wrong-answered question. Initialise calculateResult = 0

Expected Outcome:

```
Enter your name: Mark
=====
Player Name:  Mark
=====
Choose level
Enter "E" for easy, "M" for Medium, "H" for hard
Enter level e
Easy Level
Q1: Is Indent important in Python?
Your options are:
1. True
2. False
Enter your choice 3
Invalid option
|
```